MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 08** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. The distance between two parallel tangents to a circle of radius 4.5 cm is: [1]

(A) 4.5 cm (B) 9 cm (C) 13.5 cm (D) 18 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 9 cm
Justification: The distance between two parallel tangents equals the diameter of the circle: $d = 2r = 2(4.5) = \boxed{\mathbf{9}\text{ cm}}$.

Q2. In quadrilateral $ABCD$ circumscribing a circle, $AB = 7$ cm, $BC = 8.5$ cm, and $CD = 5$ cm. The length of DA is: [1]

(A) 3.5 cm (B) 4.5 cm (C) 6.5 cm (D) 5.5 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A) 3.5 cm
Justification: Since $AB + CD = BC + DA \implies 7 + 5 = 8.5 + DA \implies DA = 12 - 8.5 = \boxed{\mathbf{3.5}\text{ cm}}$.

Q3. If two concentric circles have radii 25 cm and 7 cm, the length of the chord of the larger circle touching the smaller circle is: [1]

(A) 24 cm (B) 36 cm (C) 48 cm (D) 50 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (C) 48 cm
Justification: Half-chord $= \sqrt{25^2 - 7^2} = \sqrt{576} = 24$ cm $\implies$ Chord $= 2 \times 24 = \boxed{\mathbf{48}\text{ cm}}$.

Q4. If PT is a tangent at T to a circle with centre O and $\triangle OPT$ is isosceles with $OP = \sqrt{2} \cdot OT$, then $\angle POT$ is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $90°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $45°$

Justification: In right $\triangle OTP$ ($\angle OTP = 90°$), $\cos(\angle POT) = \frac{OT}{OP} = \frac{1}{\sqrt{2}} \implies$ $\boxed{\angle \mathbf{POT = 45°}}$.

Q5. The common point of a tangent to a circle and the circle is called the: [1]

(A) Centre (B) Point of contact (C) Origin (D) Vertex

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) **Point of contact**

Justification: The unique point shared by a tangent line and a circle is called the $\boxed{\textbf{Point of contact}}$.

Q6. **Assertion (A):** The opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.
Reason (R): Tangents drawn from an external point to a circle are equal in length. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Equal tangent segments from each vertex yield SSS congruence of adjacent central triangles, proving adjacent half-angles equal and opposite central angles sum to $180°$. Both statements are true and (R) is the basis of the proof of (A).

SECTION B: Very Short Answer Type — MATHS MANTHAN | ($2 \times 2 = 4$ Marks)

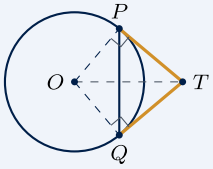
Q7. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. If $\angle PTQ = 80°$, find the measure of $\angle OPQ$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Angles in Isosceles $\triangle TPQ$ [1 Mark]
$TP = TQ \implies \angle TPQ = \angle TQP = \frac{180° - 80°}{2} = 50°$.

Step 2: Finding $\angle OPQ$ [1 Mark]
Since $OP \perp TP \implies \angle OPT = 90°$:

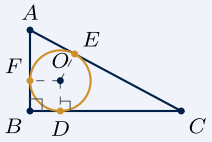
$$\angle OPQ = \angle OPT - \angle TPQ = 90° - 50° = \boxed{\mathbf{40°}}$$



Q8. Find the inradius of a right-angled triangle whose base is 15 cm and hypotenuse is 17 cm. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let right $\triangle ABC$ ($\angle B = 90°$) have base $BC = 15$ cm and hypotenuse $AC = 17$ cm.



Step 1: Third Side via Pythagoras [1 Mark]

$AB = \sqrt{AC^2 - BC^2} = \sqrt{17^2 - 15^2} = \sqrt{64} = 8$ cm.

Step 2: Inradius Calculation [1 Mark]

$r = \frac{AB+BC-AC}{2} = \frac{8+15-17}{2} = \frac{6}{2} = \boxed{\mathbf{3}\text{ cm}}$.

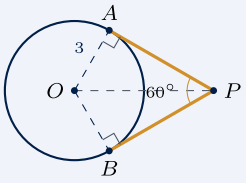
SECTION C: Short Answer Type

MATHS MANTHAN | ($2 \times 3 = 6$ Marks)

Q9. If PA and PB are tangents to a circle with centre O such that $\angle APB = 60°$ and radius $OA = 3$ cm, find the length of OP and the length of each tangent. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

OP bisects $\angle APB$, so $\angle OPA = 30°$. $\triangle OAP$ is right-angled at A.



Step 1: Finding OP [1.5 Marks]

$\sin 30° = \frac{OA}{OP} \implies \frac{1}{2} = \frac{3}{OP} \implies \mathbf{OP = 6}$ cm.

Step 2: Tangent Lengths PA and PB [1.5 Marks]

$\tan 30° = \frac{OA}{PA} \implies \frac{1}{\sqrt{3}} = \frac{3}{PA} \implies PA = PB = 3\sqrt{3}$ cm.

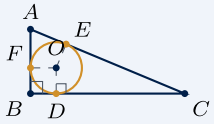
$$\boxed{OP = \mathbf{6}\text{ cm}, \quad PA = PB = \mathbf{3\sqrt{3}}\text{ cm } (\approx \mathbf{5.20}\text{ cm})}$$

Q10. A circle is inscribed in a right $\triangle ABC$ with $\angle B = 90°$, $BC = 12$ cm and $AB = 5$ cm. Find the area of the portion of the triangle that lies outside the circle. (Use $\pi = 3.14$) [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Hypotenuse & Inradius [1.5 Marks]

$AC = \sqrt{5^2 + 12^2} = 13$ cm. Thus $r = \frac{AB+BC-AC}{2} = \frac{5+12-13}{2} = 2$ cm.



Step 2: Required Area Outside Circle [1.5 Marks]

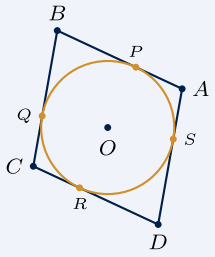
Area($\triangle ABC$) $= \frac{1}{2} \times 5 \times 12 = 30$ cm^2; Area(Circle) $= 3.14 \times 2^2 = 12.56$ cm^2.

$$\boxed{\text{Required Area} = 30 - 12.56 = \mathbf{17.44}\text{ cm}^2}$$

Q11. Prove that the parallelogram circumscribing a circle is a rhombus. Hence, if one of the interior angles of this circumscribing parallelogram is 90°, prove that it must be a square. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Parallelogram $ABCD$ circumscribes a circle touching AB, BC, CD, DA at P, Q, R, S.



Part 1: Proving $ABCD$ is a Rhombus [3 Marks]

1. By Theorem 10.2: $AP = AS$, $BP = BQ$, $CR = CQ$, $DR = DS$.
2. Adding: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ) \implies AB + CD = AD + BC$.
3. Since $ABCD$ is a parallelogram, $AB = CD$ and $AD = BC \implies 2AB = 2BC \implies AB = BC$.
4. Hence $AB = BC = CD = DA$, so $ABCD$ is a **rhombus**. [Proved]

Part 2: Proving $ABCD$ is a Square when One Angle is $90°$ [2 Marks]

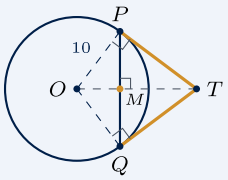
Suppose $\angle A = 90°$. Since $ABCD$ is a parallelogram, opposite angles are equal ($\angle C = 90°$) and adjacent angles are supplementary ($\angle B = \angle D = 180° - 90° = 90°$). As all four sides are equal and all four interior angles equal 90°,

$$\therefore ABCD \text{ must be a square} \quad \text{[Proved]}$$

Q12. PQ is a chord of length 16 cm of a circle of radius 10 cm. The tangents at P and Q intersect at a point T. Find the length of the tangent TP and the area of $\triangle TPQ$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Join OT meeting chord PQ perpendicularly at midpoint M. So $PM = MQ = 8$ cm.



Step 1: Finding OM [1 Mark]

In right $\triangle OPM$: $OM = \sqrt{OP^2 - PM^2} = \sqrt{10^2 - 8^2} = 6$ cm.

Step 2: Similarity $\triangle OPT \sim \triangle PMO$ for TP [2 Marks]

$\angle OPT = \angle PMO = 90°$ and $\angle POT = \angle MOP$. By AA similarity:

$$\frac{TP}{PM} = \frac{OP}{OM} \implies \frac{TP}{8} = \frac{10}{6} \implies \mathbf{TP = \frac{40}{3}} \text{ cm}$$

Step 3: Altitude TM and Area of $\triangle TPQ$ [2 Marks]

In right $\triangle PMT$: $TM = \sqrt{TP^2 - PM^2} = \sqrt{\left(\frac{40}{3}\right)^2 - 8^2} = \sqrt{\frac{1024}{9}} = \frac{32}{3}$ cm.

Area$(\triangle TPQ) = \frac{1}{2} \times PQ \times TM = \frac{1}{2} \times 16 \times \frac{32}{3} = \frac{256}{3}$ cm^2.

$$\boxed{TP = \mathbf{\frac{40}{3}} \text{ cm}, \quad \text{Area}(\triangle TPQ) = \mathbf{\frac{256}{3}} \text{ cm}^2 \ \left(\approx \mathbf{85.33} \text{ cm}^2\right)}$$

Q13. Laser Alignment across Circular Optical Cavities

In an optical laboratory, a laser beam emitter at point P directs two beams PA and PB tangent to a circular optical resonator of radius $r = 3$ cm centred at O. The emitter is placed at distance $OP = 6$ cm.

(i) Find the angle $\angle APO$ between the emitter line and tangent beam PA. [1]

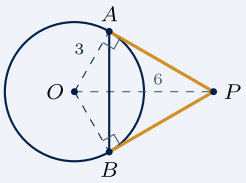
(ii) Calculate the total angle of beam divergence $\angle APB$. [1]

(iii) Find the length of each laser beam PA, OR determine the length of the chord of contact AB. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

(i) Angle $\angle APO$: [1 Mark]

In right $\triangle OAP$: $\sin(\angle APO) = \frac{OA}{OP} = \frac{3}{6} = \frac{1}{2} \implies$ $\boxed{\angle \mathbf{APO} = \mathbf{30^\circ}}$.



(ii) Total Beam Divergence $\angle APB$: [1 Mark]

Since $\triangle OAP \cong \triangle OBP$, $\angle APB = 2\angle APO = 2(30^\circ) = \boxed{\mathbf{60^\circ}}$.

(iii) Main Choice — Length of Each Laser Beam PA: [2 Marks]

In right $\triangle OAP$: $PA = \sqrt{OP^2 - OA^2} = \sqrt{6^2 - 3^2} = \sqrt{27} = \boxed{\mathbf{3\sqrt{3}}\text{ cm} \quad (\approx \mathbf{5.20}\text{ cm})}$.

— OR — (Alternative Choice — Length of Chord of Contact AB): [2 Marks]

Since $PA = PB = 3\sqrt{3}$ cm and $\angle APB = 60^\circ$, $\triangle PAB$ is equilateral:

$$\boxed{\text{Chord } AB = PA = \mathbf{3\sqrt{3}}\text{ cm} \quad (\approx \mathbf{5.20}\text{ cm})}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★